



SolarFlow PM

COMPLETE PROJECT DOCUMENTATION

Solar project management & CRM platform

Every module, every automation, the AI foundation, what has been built to date, what is still open, and the roadmap to a pipeline that runs itself — with solar and CRM best practice alongside.

Platform

Architecture, roles, security, database, deployment

26 modules

Delivery pipeline, portals, mobile app, CRM, signing

Automation & AI

What runs today, and the path to zero-touch

Status

85 commits of work, known gaps, open items

Best practice

Solar operations, CRM, security, KPIs

Appendices

Migrations, screens, APIs, tests, glossary

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How to read this document

Chapters 1–4 describe the platform as a whole. Chapters 5–9 are the module catalogue: every module, what it does, the screens, the data, and the automation inside it, each tagged **LIVE**, **PARTIAL**, **PLANNED** or **GAP**. Chapters 10–14 cover the end-to-end flow, the automation and AI running today, the work done and what is still open. Chapters 15–17 are the roadmap to a pipeline with no routine human steps. Chapters 18–21 are best practice and the numbers to run the business by.

Executive summary

SolarFlow PM is one system for the whole life of a residential solar sale, from the first contact to a completed installation. Office staff, sales reps, dealers, designers, finance and the homeowner all work in it.

26

MODULES

7

USER ROLES

53

SCREENS

64

API ENDPOINTS

88

DATABASE TABLES

190

SECURITY POLICIES

43

SCHEMA MIGRATIONS

85

COMMITTS SHIPPED

What it does today

- **Sells.** A CRM of every contact, with duplicate detection across every email and phone on file, and a Contact stages board from first contact to signed or lost.
- **Signs.** Contract signed records the system sold and creates the project in one step; the project then holds the contact in place.
- **Delivers.** A seven-stage pipeline (Survey to Complete, plus Hold and Cancelled) with a form per stage; a project cannot advance with paperwork missing.
- **Serves the homeowner.** A plain-language portal and iOS/Android app: current stage and its typical duration, chat with their PM, uploads, push, and a rating after every stage.
- **Serves dealers and measures.** A dealer portal (own projects, leads, commissions), a live dashboard, and a report builder with Excel/CSV export.

Where it stands

The delivery pipeline, portals, mobile app, chat, feedback, dashboard, reports and the core CRM are built and covered by automated tests. The platform runs on plain PostgreSQL (Neon) and Vercel. The application applies its own database migrations from **Admin** → **Database**, so no SQL console is needed.

The honest position on AI

There are no AI model calls in the product yet. The foundation is in place: tables built for AI quoting, booking, adders and vendor quotes, and an exceptions queue for low-confidence AI output. Chapters 12, 16 and 17 set out what to build on it.

The three most valuable next steps

1. **Turn on the automation that is already written.** Install reminders, quiet-hours chat delivery and the unanswered-message digest run from one endpoint, but no schedule calls it. Adding a cron entry is a one-line change (chapter 14).
2. **Automate the paperwork bottleneck with AI.** Utility-bill and document extraction, permit-packet assembly, and permit/interconnection status tracking remove the most repetitive office work, and the stage forms already have typed columns ready for the answers (chapter 16).
3. **Close the loop on speed-to-lead.** Instant, automated first response and appointment booking on every new contact. This is the single largest driver of close rate in residential solar (chapter 19).

The platform at a glance

Five groups of modules, one database, one API, and one set of access rules, shared by the web app, the portals and the mobile app. Nothing is copied or synced between them.

GROUP	MODULES	STATUS
Platform	M01 Database & self-applying migrations · M02 Authentication & sign-in · M03 Roles & row-level security · M04 Files & documents · M05 Audit & activity log	LIVE
Project delivery	M06 Pipeline board & stage engine · M07 Stage forms · M08 Hold, Cancel & Complete · M09 Projects & project details · M10 Admin panel & reference lists · M11 Analytics dashboard · M12 Report builder	LIVE
Dealers & partners	M13 Dealer portal · M14 Dealer companies & commissions · M15 Lead review queue	LIVE
Homeowner experience	M16 Customer portal · M17 Customer management · M18 Customer mobile app · M19 Project chat & inbox · M20 Stage feedback & follow-ups	LIVE
CRM & sales	M21 CRM foundation · M22 Contacts & Create Contact · M23 Contact stages · M24 Contract signing → project · M25 Deals · M26 Subscribers	LIVE M26 PARTIAL

Channels

Web application Staff, sales, dealers, designers and finance, each landing on the screen for their role. Deployed on Vercel from the <code>main</code> branch.	Customer portal The homeowner's view in any browser: a mobile-first, installable web app with dark mode, offline support and push.	iOS & Android apps Native shells (Capacitor) around the same portal, built from GitHub Actions with no Mac or Android Studio needed. App ID <code>com.integratesun.solarflow</code> .
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Scale of the codebase

MEASURE	VALUE	MEASURE	VALUE
Application code (TypeScript)	41,200 lines	Database code (SQL migrations)	12,300 lines
Screens	53	API endpoints	64
Database tables	88	Database functions	118
Row-level security policies	190	Database triggers (named)	19
End-to-end test suites	9 (incl. 1 in a real browser)	SQL verification suites	2 (RLS matrix + auth flows)
Unit tests	62	Commits, 3 Aug – 22 Sep 2026	85

Architecture & technology

A deliberately small stack. The database carries the rules that matter: who can see what, what a stage move requires, and what signing does. So every screen, the portals, the app and any future automation get the same answers.

Stack

LAYER	TECHNOLOGY	WHY
Web framework	Next.js 15.5 (App Router), React 19, strict TypeScript	Server components for data-heavy screens; client islands only where there is interaction (boards, forms, dialogs).
Database	PostgreSQL 17 on Neon (serverless, pooled endpoint); driver <code>pg</code> 8	Plain Postgres, no vendor lock-in: the platform replaced Supabase with its own auth engine and file storage inside the database.
Access control	Row-level security on every table; per-request identity claims	Access rules live in one place and cannot be bypassed by a new screen or a forgotten filter.
Authentication	In-database engine: <code>bcrypt</code> (cost 12), opaque 256-bit session tokens, SHA-256 at rest	7-day revocable sessions; deactivating a user ends access on their next request.
Files	File bytes stored in Postgres, behind two security-definer functions	Right-sized for site photos and plan PDFs; metadata already shaped for S3 if volume grows.
Email	SMTP via Nodemailer	Invites, password recovery, chat notifications, rating emails, digests.
Push	Web Push (VAPID) + Capacitor push for native	Install reminders, stage updates, PM asks and chat, with quiet hours.
Spreadsheets	ExcelJS	Report-builder exports to <code>.xlsx</code> and <code>.csv</code> .
Mobile	Capacitor 8 (camera, push, preferences, share)	One codebase for iOS and Android, built in GitHub Actions.
Hosting	Vercel (Node runtime, 60-second functions); GitHub for source and CI	Push-to-deploy from <code>main</code> ; previews disabled for the working branch.

Design principles the code follows

One source of truth

The Deals board is the Pipeline's own board over the projects deals became, not a copy. The contact's stage lives on the person. Field lists such as the intake registry are data, and they drive the forms, the save allowlists and the signing popup together.

Rules in the database

A won deal cannot be un-won, a signed contact with a project cannot be moved, a customer cannot read an internal chat note, and a dealer cannot see another dealer's project. All of this is enforced by triggers, security-definer functions and policies, not by the UI.

Atomic business steps

Signing records the system, moves the contact and creates the project in one statement. Deal-to-project conversion carries the documents across. A refusal leaves nothing half-written.

Degrade, never break

If the database is a migration behind, screens show which file is missing rather than crashing. `/api/health` reports the build, the database endpoint and every migration.

Database migrations

The schema is 43 ordered migration files (appendix A). Each is idempotent and guarded: it refuses, with the name of the file to run first, if a prerequisite is missing. The deployment carries its own migration files, and **Admin** → **Database** shows which are applied and applies the rest with one button, each file in its own transaction, recording what ran. Detection works by probing for what each file creates, including the shape of a function where the name alone is ambiguous.

Lesson built in

A migration that has shipped is never edited again; a correction always goes in a new file. The one time an applied file was rewritten, databases that already had it could not see the change, which is why the signing function now lives in its own file (004100) and is detected by what it returns.

Users, roles & access

Seven roles. Each lands on its own home screen, and each sees exactly what the database's row-level security allows. Hiding a button is never the protection.

ROLE	LANDS ON	CAN SEE AND DO
Admin	Pipeline	Everything, including user management, the activity log, the finance view, database migrations, moving projects backwards, reinstating cancelled projects and deleting projects.
Ops / PM	Pipeline	All project data: runs the pipeline, fills stage forms, manages customers, contacts and deals. Not user management, the audit log or the finance view.
Sales	Deals	Contacts, Contact stages and deals they own, the unassigned pool, or all deals, depending on the capability or profile setting. Sees the projects of deals they can see, view-only.
Designer	Designer queue	Projects where they are the assigned designer, plus the internal design tables for those projects.
Finance	Finance	Whitelisted financial columns through a dedicated view, plus price book, change orders, vendor quotes and adders. No direct project or client rows.
Dealer	Dealer dashboard	Their own company's projects (read-only, with per-field visibility), lead submission, commissions, their own settings.
Customer	Portal	Their own project in plain language, customer-facing documents and photos, chat with their PM (never internal notes), requests, ratings.

Sign-in

Three front doors: **staff**, **dealer** and **homeowner**. Behind them is one login function, one validation path, one rate limiter and one role router. Ten failed passwords lock an account for ten minutes. Password recovery and invitations use 7-day, single-use links. Admins can set a password with force-change-on-first-login. Homeowners are auto-invited when their project is created.

Access that needs no login

LINK	GRANTS
<code>/u/<token></code>	Uploading photos to one project, for a crew member or homeowner without an account. The token is hashed at rest, expires within 7 days and can be revoked, and no session is ever created.
<code>/r/<token></code>	Recording one stage rating from the email's five faces. The score is saved before the page renders, because requiring a login loses most responses.

Platform modules

The layer everything else stands on: the database and how it keeps itself current, who can sign in, what each role may see, where files live, and the record of who did what.

M01 Database & self-applying migrations

LIVE

Migrations 000000–000400, db-migrations/db-apply, Admin → Database, /api/health

Plain PostgreSQL with no vendor-specific features, so it runs on Neon, any managed Postgres or a local install. The app role connects through a pooled connection and switches to an `authenticated` role per request, carrying the user's identity as claims that the security policies read.

Screens	Admin → Database: every migration the deployment carries, which are present, and one <i>Apply</i> button that runs the missing ones in order, each in its own transaction, recording them in <code>schema_migrations</code> and reporting PostgreSQL's exact words on any refusal.
Health	<code>/api/health</code> : build commit, which environment variables are set, a masked database endpoint (so two deployments can be compared), a branch marker, the database connection result and the list of migrations the database is behind.
Safety	Every migration is idempotent and guarded. When a module is missing, screens show a notice naming the file rather than failing, and errors name the real cause. SQL bundles in <code>db/dist</code> remain for console use.

M02 Authentication & sign-in

LIVE

Migrations 001000, 002600, 002700, 003000 · docs/auth.md

A complete auth engine inside the database, replacing Supabase Auth. Credentials, sessions and one-time tokens are reachable only through security-definer functions; the application role has no direct access to the tables.

- bcrypt passwords (cost 12); sessions are opaque 256-bit tokens in an httpOnly cookie, with only the SHA-256 stored; 7-day lifetime, revocable server-side.
- Lockout after 10 consecutive failures, for 10 minutes. One sign-in function behind three doors (staff, dealer, homeowner), with role-based routing.
- Invitations and password recovery by 7-day single-use links, for every role including homeowners. Admin-set passwords can force a change at first login. Every password field has a show/hide control.

M03 Roles & row-level security

LIVE

Migrations 000300, 000600, 000800, 000900, 003300, 004200 · docs/rls-matrix.md

190 policies across 88 tables implement the role contract in chapter 4. Functions lose the default public execute right, and sensitive writers run as security-definer with explicit role checks. The finance role reads only whitelisted columns through a view. Sales reps see a deal's project exactly when they can see the deal: the policy asks the deals table rather than repeating its rules.

M04 Files & documents

LIVE

Migrations 000700, 001100 · lib/storage, /api/files, /api/documents, /u/<token>

Private buckets (drawings, PDF deliverables, photos) with the bytes stored beside their metadata in Postgres. Documents are filed against a deal before signing and gain the project link at conversion, so nothing is uploaded twice. Customers see only customer-visible documents. Crews and homeowners can upload photos through time-limited, revocable links without an account.

M05 Audit & activity log

LIVE

Migration 000500 (+ kinds, deal/client links in 003400) · Admin → Activity log

One shared writer that every module calls. It records actor, role, action, entity, context, the kind of activity (call, email, SMS, meeting, note, form, stage move, field change, login, system) and links to project, deal and client. Rows cannot be edited. Stage moves, signings, project deletions, merges and admin actions are all logged; password values never are. The same log is the contact's Activity tab and the deal timeline.

Project delivery modules

From a signed contract to a system producing power: the pipeline the project team runs every day, the forms that record each step, and the numbers that show how it is going.

M06 Pipeline board & stage engine

LIVE

Migration 001200 · `/pipeline, /projects, lib/stages · moves via /api/projects/[id]/move`

1 Survey Site survey	2 Design Plans & stamps	3 Permits Permit · ICA · HOA · NTP	4 Procurement Material	5 Install Installation	6 Inspection & PTO Inspection, PTO, energization	7 Complete Terminal	Hold · Cancelled Side stages
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- **Forward, one stage at a time**, running the same validation as the stage form's Advance button. A rejected drop snaps back with a red outline and a toast listing exactly what is missing, linked to the stage form.
- **Backwards** is admin-only, one stage, with a logged reason, including back out of Complete, which reopens the project.
- **Cards** show system size, days in stage, the missing-items count, unread customer messages and open low-rating follow-ups.
- **Stage history** is written by a trigger on every stage change, and it feeds "days in stage", the dashboard's ageing and typical durations.
- **Live**: boards re-read every 10 seconds while visible and on focus, and immediately when another tab announces a move.
- Only the PM (ops) or an admin moves projects. The Projects tab is the same data as a filterable table.

M07 Stage forms

LIVE

Migration 001400 · `/projects/[id]/stages/[stage] · lib/stages/fields.ts`

A data-entry form per stage, with typed columns for every field, so later automation needs no schema work. "Days" fields are computed from dates at read time, never stored. Every stage closes on a **Drive Updated** gate.

STAGE	WHAT IT RECORDS
Survey	Site survey status, completed date, days, adders details.
Design	Designer, design status, requested/received dates, LightReach shading & monthly production-loss report and date, PM notes; <i>Stamps</i> : status, requested/received dates.
Permits	Five tracks, each with status, applied/received dates, notes and revision notes: Building permit (required permits: building / electrical / structural), ICA (interconnection agreement), HOA , HDM NTP (notice to proceed: submitted/approved) and the Cash M2 milestone.
Procurement	Procurement manager, material status, requested/delivered dates, PM notes.
Install	Install manager, status, requested/scheduled/completed dates, install pictures.
Inspection & PTO	Inspection status, failed-inspection notes, requested/completed dates; PTO status, applied/received dates; monitoring / system energization status and date.
Complete	Total project days, completion status and date, completion notes, final Drive Updated.

STAGE	WHAT IT RECORDS
Payments	Down payment and Cash M1-M3 (not requested → requested → initiated → received); Finance M1/M2 labelled by partner (not submitted → submitted → approved / rejected), stored once per project.

M08 Hold, Cancel & Complete

LIVE

Migrations 001500, 001600

Any project can go on Hold or be Cancelled from any stage, bypassing validation, because a customer pulling out must never be blocked. The move needs a reason from a list and notes. Hold reasons: customer request, finance pending, weather, access issue, awaiting documents, site issue, other, with an expected resume date. Cancellation reasons: customer cancelled, failed credit, site not viable, permit denied, duplicate, competitor, other, with refund/clawback and equipment-return flags. The origin stage is stored, so Resume and Reinstate (admin-only) restore the project exactly. Held and cancelled projects leave active counts but are never deleted.

M09 Projects & project details

LIVE

Migrations 001700, 001800, 002100 · </projects/new>, [/projects/\[id\]](/projects/[id])

The New Project form and the project record: customer, site address, dealer, sales rep, jurisdiction, utility, and a system specification (system type, module type and quantity, inverter type and quantity, battery type and quantity, system size) plus financing (cash or financing, company, notes, estimated completion shown to the customer). Equipment and financing values come from admin-managed lists stored by ID, so fixing a typo in a module name fixes it everywhere. The project page has panels for customer, details, system specification, financing and commission, a stepper to each stage form, and the header actions (hold, cancel, resume, reinstate, delete).

M10 Admin panel & reference lists

LIVE

Migration 001300 · /admin/*

AREA	CONTENTS
People & teams	Users & roles (invite, resend, cancel, set password, force change, deactivate, guarded delete), surveyors, designers, install crews, vendors, dealers, contacts, sales reps.
CRM lists	Sources, loss reasons, roof types, competitors, dealer tiers.
Equipment & financing	System types, module types, inverter types, battery types, financing companies, cash or financing, finance partners, dealer visibility.
Operations	Canned replies (chat), company settings, activity log, Database (migrations).

Lists are deactivated, not deleted, so history is never broken.

M11 Analytics dashboard

LIVE

Migration 002800 · </dashboard> (admin, ops, finance)

One full-width scrolling page in four bands, densest at the top. Nothing is a hand-maintained list: every figure is a live aggregate grouped by PM, dealer and stage, so a new hire's projects appear the first time one is assigned. Stat cards stream first and charts fill in behind them. Per-stage ageing thresholds live in the database (`stage_thresholds`) so they can be re-tuned without a release. A quarter is the default period, sample sizes are shown plainly, and a count and its share are shown as two separate figures.

M12 Report builder

LIVE

SCHEDULES

Migration 002200 · /reports, /reports/builder

Drag-and-drop reports opened blank, from a template or from a saved report. A report is a JSON definition, never SQL: the generator builds a parameterised query from whitelisted columns, so nothing a user writes is ever executed. The field library is filtered to what the role may see. Output is preview, print, Excel (.xlsx) or CSV. Reports can be private, shared with a role or shared with named users, and every run is recorded.

Open item

The data model stores report **schedules**, but nothing yet runs them on a timetable. See chapter 14.

Dealer & partner modules

Dealers bring sales. They see their own book, submit leads and follow their commissions, all read-only over data the project team enters, and scoped in the database so one dealer can never see another's.

M13 Dealer portal

LIVE

Migration 001900 · /dealers, /dealers/projects, /dealers/leads, /dealers/commissions · sign-in door: dealer

- **Dashboard:** four stat cards, projects by stage, recent activity ("what is moving") and needs attention ("what is stuck").
- **Projects:** their company's projects and each project's detail, with **per-field dealer visibility** set by the admin, so a dealer sees exactly the fields the business chooses to show.
- **Leads:** the dealer submits a lead into a queue the PM reviews. It never creates a project directly. The dealer sees the status and the resulting project once converted.
- **Commissions:** their commission lines per project.
- The only dealer write paths are lead submission and their own account settings.

M14 Dealer companies & commissions

LIVE

Migration 002000 · Admin → Dealers · /api/projects/[id]/commission

Each dealer company has a full record covering identity, commercial terms, portal behaviour and **commission defaults**. A trigger seeds a new project's commission from those defaults. Names are unique regardless of case and spacing (so "SunPro Energy LLC" and "sunpro energy llc " are caught at creation), and every change is audited. A company with projects or leads cannot be deleted by any code path, because the foreign keys refuse it. Dealer users are linked to companies, and a company can be created inline from the project form.

M15 Lead review queue

LIVE

/leads (admin, ops) · /api/leads

Everything dealers have submitted, with Review, Convert and Decline actions. Converting creates the client and the project prefilled from the lead, links it back for the dealer, and **auto-invites the homeowner** with a set-password link. Since the CRM foundation, dealer submissions also live on as deals, keeping their attribution. The original `Leads` view still answers for anything that reads it.

Homeowner experience modules

The homeowner is kept informed without having to phone: where their project is, how long this part usually takes, what is needed from them, and a direct line to their project manager.

M16 Customer portal

LIVE

Migrations 002300, 003100 · `/portal`, `/portal/project`, `/portal/documents`, `/portal/photos`, `/portal/messages`, `/portal/more`

- **Shows the stage they are in, not the seven they could be in:** one clear card with progress, the **typical duration** of the current stage ("Typical 15–30 days") and what is up next ("Up next: Design · 7–10 days"). This answers the question behind most status calls before anyone picks up the phone.
- **Plain language:** one admin-editable mapping layer turns internal values ("ICA Status: Applied") into homeowner wording. A raw dropdown value is never shown.
- **Four genuine actions** land as requests in the PM's queue rather than writing stage data: give availability, ask a question, update contact details, send a document.
- An optional **PM-set completion estimate** (no estimate is better than a wrong one), a "call my PM" card, and customer-visible documents and photos with ZIP download.
- Typical durations are configuration, not code: they are seeded from the business's own history once enough projects have completed.
- Dark mode, installable, mobile-first.

M17 Customer management

LIVE

Migration 002400 · `/admin/customers`, `/admin/people`, `/api/customers/*`

The customer is a person and the project is a job: one client can carry several projects. This module finds, corrects, merges and controls portal access for existing records. It covers search, merging duplicates with a field-by-field choice, archiving, and **anonymising** for data-deletion requests. Portal access is managed here too: send an invite, set or reset a password, revoke. A customer with projects or leads cannot be deleted, and destructive operations run through admin-checked database functions.

M18 Customer mobile app (iOS & Android)

LIVE

Migration 002500 · Capacitor projects in `/android`, `/ios` · GitHub Actions: Build Android app, Build iOS app · `docs/mobile-store-submission.md`

- An installable portal with six tabs (Home, Messages, Project, Docs, Photos, More). It adds camera upload, offline support and native push. It uses one database, one API and one set of security rules, with nothing copied or synced.
- **Push** for stage changes, PM asks, chat and **install reminders 48 and 24 hours before**. It respects per-customer notification preferences, and every delivery is recorded, keeping pushes under ten per project in a measurable way.
- **PM asks:** the PM requests something (a photo of the electrical panel, a signature, a date) and the customer answers in the app.
- The store app contains the customer portal and nothing else. Builds refuse a pinned deployment URL (which would never update), and in-app account deletion meets store rules.

- A debug APK is built from the GitHub website with no signing setup. Signed release bundles are built when the signing secrets are present.

M19 Project chat & Messages inbox

LIVE

Migration 002900 · [/projects/\[id\]/chat](#) (+ print view), [/messages](#), [/portal/messages](#)

- One thread per project between the PM and that project's customer, kept on the project record for the life of the job.
- **Internal notes are walled off in the database:** a customer's query cannot return one, and a customer cannot create one. In the UI they are a separate tab in a different colour.
- **Messages** is the PM's global inbox across all projects, unread first then newest, so a message from three days ago is never buried.
- Canned replies, flags, read tracking, unread badges on pipeline cards, and a print view for the record.
- **Notifications:** push if the customer has the app, otherwise email. Quiet hours hold messages between 9pm and 8am local and deliver them in the morning rather than dropping them. PMs get a per-message email if they opt in, otherwise a **twice-daily digest** of unanswered messages.

M20 Stage feedback & follow-ups

LIVE

Migration 003200 · in-portal sheet, email faces via [/r/<token>](#), [/tasks](#) (Follow-ups), [/feedback](#) (verbatim log)

- A **one-tap rating the moment each stage completes**, six or seven times per project instead of one survey at the end, so you learn *which stage* lost the customer while it can still be fixed.
- The score is saved on tap. Reasons ("chips") and a comment are optional and attached separately. Every score offers the comment box.
- **Email fallback:** if a rating is not answered in the app within 24 hours, one email with five clickable faces goes out. Clicking records the score immediately, with no login.
- A **low score opens a follow-up task** (Follow-ups, oldest first) that stays on the pipeline card until someone closes it with a note.
- The **verbatim log** lists every comment with score, stage, date and chips, filterable and searchable.

CRM & sales modules

The half of the business that happens before anybody signs: every person you have ever spoken to, where each one stands, the moment they sign, and the sale followed through to a finished install.

M21 CRM foundation

LIVE

Migrations 003300, 003400 · Sales role · the People model

- **One person, one record:** `clients` is the spine. A person moves from prospect to customer with no copying: their lifecycle (lead, prospect, customer, past customer) is derived from the projects that exist, never typed by hand.
- **Channels:** every email and phone a person has, normalised and matched, with one primary per kind kept in sync with the person's main columns. The duplicate check searches all of them, so a second address is still the same person.
- **Addresses:** mailing and property addresses kept separately, so a person with two houses has two deals rather than two records.
- **Sales role** with deal visibility (own, team, all), a "manage all deals" capability, and the unassigned pool every rep can work.
- Activity kinds on the audit log (call, email, SMS, meeting, note, form...) linked to deal and client, so a logged call "touches" the person.
- Dealer submissions (the old leads table) became deals with their attribution and a stage, and a compatibility view keeps old readers working.

M22 Contacts & Create Contact

LIVE

Migrations 003600, 003700 · `/admin/people`, `/admin/people/new`, `/admin/people/[id]` · `lib/crm/intake.ts`

Contacts holds **everybody**, customers or not: a simple list with name, lifecycle, email, phone, city/state and last activity, defaulting to "Everyone". **Create Contact** is a full page asking only for the 24 contact fields:

GROUP	FIELDS
Contact	Contact owner (auto-set to whoever is creating), salutation, first name, last name*, email / secondary email, phone / mobile*, owner's phone, lead source (set once), dealer, consultant, created by (recorded), description.
Mailing address	Street, city, state, ZIP, country.
Status	Lead status (the contact stage), lost reason, reschedule reason.
Dealer & notes	Dealer code, Wave sales notes, additional information.

* Needed to save: a last name, and an email or a phone. A rep on a driveway with only a mobile number can still save.

- **Duplicate guard** across every channel on file, offering to open the existing record, with "they are a different person — save anyway". **Save and New** for data entry.
- Filling a deal-side field (dealer code, notes, reschedule reason) creates the first deal alongside the contact in the same statement. A business-card contact gets no empty deal.
- The **contact record** opens as a full page with pill tabs: Details, Contact details, *System* (only once a system has been recorded at signing), Projects, Deals, Subscriptions, Portal access, Activity.

- The system, usage and price fields are not on the contact. They live on the deal, and one registry drives the forms, the save allowlists and the signing popup, so the lists cannot drift apart.

M23 Contact stages

LIVE

Migration 003800 · /admin/people/stages

Contact created On file, not booked	Appointment scheduled A date in the diary	Appointment rescheduled Moved at least once	No-show Nobody there	Quoted A number is with them	Financing approved The money is in place	Contract signed Signed	Lost Closed, reversible
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- The stage belongs to the person, so every contact has exactly one card from the moment they exist. It is the same field as *Lead status* on the record.
- **Any stage to any stage**, because diary states move sideways and backwards (a no-show goes back to rescheduled). The one exception is Contract signed, which is a step (M24).
- Optimistic drag with snap-back on refusal, search, "only assigned", days in stage, last contact. The board fills the window with all eight columns on a 1440px screen. **Lost is red** and Contract signed is green.
- Every move is written to the activity log with its direction. The clock restarts only when the stage changes.

M24 Contract signing → project

LIVE

Migrations 003900, 004000, 004100 · public.sign_contact, public.delete_project · /api/contacts/[id]/sign, DELETE /api/projects/[id]

Drop on Contract signed or choose it in Lead status	Signing form Dealer*, site address*, system size*, modules, inverter, battery, usage, price, financing	One transaction System recorded on the deal · contact signed · project created at Survey	Held Contact locked in Contract signed while the project exists
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- **Signing is a step, not a drag.** Nothing moves until the form is saved, and Cancel leaves the contact where they were. The form opens pre-filled from their open deal (a 5.4 kW quote arrives as 5.4) and from their dealer and mailing address.
- **The project is created automatically** through the same conversion the deal board uses: system size, brands, battery, module count and contract value are copied; the deal's documents gain the project link; the deal becomes Won. A contact with no deal gets one, at Contract out. Any refusal leaves nothing behind.
- **The hold:** a database trigger refuses every path out of Contract signed (board, Lead status, SQL) while the project exists. The board shows the card locked with its project link, and the record explains why.
- **Delete project** (admin only, with the project code typed to confirm) unwinds the sale. The project's stages, tasks, messages and forms are removed. The deal is reopened at Contract out with its system, the deal's documents such as the signed agreement are kept, and the activity log is kept. The contact is then free to move.
- Only the system, usage, money, dealer and address columns can be written by signing, whatever the request carries, and whole-number fields are rounded instead of failing.

M25 Deals

LIVE

Migration 003500, 004200 · /deals, /deals/new, /deals/[id] · lib/deals

- **Board:** the project stages Survey → Complete with Hold and Cancelled. It is the Pipeline's own board over the projects deals became, so the two can never disagree. It updates **live**: a change in another tab arrives in under a second, and anyone else's within 10 seconds, with no reload. A note points to deals not yet signed.
- **Sales reps** follow their sales here view-only (the project team moves projects), and their cards open the deal.
- **Table:** every deal, signed or not, in the deal's own stages (New → Contacted → Qualified → Proposal → Negotiation → Contract out → Won / Lost), with probability (computed or set by hand), next action and due date, value, weighted pipeline and missing items.
- **Deal record:** qualification, logged activity (a voicemail is an attempt and does not move the deal; a real conversation with a next action does), versioned proposals (the newest sets the price), Solar details, documents.
- **Rules in the database:** New cannot advance without two-way contact; a skip is allowed only when every skipped stage is satisfied; Won requires a signed contract and creates the project; a won deal cannot be un-won (except by an admin deleting its project); Lost needs a reason from the list and can be reopened.

M26 Subscribers

PARTIAL

/admin/subscribers · E-book subscribers

The subscriber list, with the **consent basis always visible**, because that is the record produced if someone complains. Still to build: the sender, double opt-in confirmation and the preference page (chapter 14).

The end-to-end journey

One homeowner, from the first contact to a system producing power. Each step below is either done by the system on its own, or done by a person with the system checking the work.

Sale

1 · Contact arrives Create Contact, a dealer lead, or a web form. Duplicates caught on every channel; owner set to the creator.	2 · Appointment Scheduled, rescheduled or no-show on Contact stages; every move logged.	3 · Quote & finance Deal with versioned proposals; financing approved.	4 · Contract signed Signing form, then system recorded, project created at Survey, contact held.
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Delivery

5 · Survey Site survey; homeowner auto-invited; rating asked on completion.	6 · Design Plans, shading report, engineering stamps.	7 · Permits Building permit, ICA, HOA, NTP in parallel tracks.	8 · Procurement Material ordered and delivered.	9 · Install Scheduled; 48 h and 24 h reminders; photos uploaded.	10 · Inspection & PTO Inspection passed; utility PTO; "power on" push.	11 · Complete Total days, final Drive Updated, last rating.
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At every step: the homeowner sees the current stage in plain language with its typical duration; the PM sees cards ageing against thresholds; the dealer sees the project move in their portal; each stage change is written to history and the activity log; and the dashboard and reports reflect it immediately.

Who touches what

STEP	PERSON	SYSTEM
Contact capture	Rep or dealer types it	Dedupes, normalises channels, assigns owner, logs.
Appointment	Rep books and moves the card	Stamps the time in stage, logs the move.
Signing	Rep fills the system form	Creates the deal if needed, the project, the document links; holds the contact.
Each stage	PM fills the stage form	Validates, blocks incomplete advances, computes days, notifies homeowner, asks for a rating.
Customer contact	PM answers chat and asks	Push or email, quiet hours, digest, unread badges.
Unhappy customer	PM closes the follow-up	Low score opens a task and flags the card until it is closed.

Automations running today

Every automatic behaviour in the product, grouped by what triggers it. Those marked **NOT SCHEDULED** are written and tested but only run when started by hand until a schedule is added.

Business workflows

WHEN...	...THE SYSTEM AUTOMATICALLY
A contact is created	Normalises and indexes every email and phone; checks all channels for duplicates; sets the contact owner to the creator; stamps who created it; creates the first deal only if deal fields were filled; composes the mailing line; starts the stage clock.
A contact changes stage	Restarts the days-in-stage clock (only on a real change); writes the move and its direction to the activity log.
A contact is signed	Creates a deal if there is none; writes the system onto it; moves the contact; converts the deal into a project at Survey with system, brands, quantities and contract value; links the deal's documents to the project; marks the deal Won; puts the dealer on the contact; locks the contact in Contract signed.
A deal is won (deal board)	Refuses without a signed contract; creates the project in the same transaction; carries contract, specification and documents; flips the person to customer.
A dealer lead is converted	Creates the client and project prefilled from the lead; links it back for the dealer; invites the homeowner.
A project is created	Seeds commission from the dealer's defaults; auto-invites the homeowner with a 7-day set-password link; fixes the originating dealer so reporting cannot drift.
A project moves stage	Validates the stage form and refuses with the exact missing items; records stage history; pushes "stage advanced" (and "power on" at energization) to the homeowner; refreshes every open board (instantly in the same browser, within 10 s elsewhere).
A project goes on hold	Requires a reason; stores the origin stage for an exact resume; notifies the homeowner; removes it from active counts.
A stage completes	Asks the homeowner for a one-tap rating; after 24 hours without an answer, emails five clickable faces; on a low score, opens a follow-up task, flags the pipeline card and alerts the PM.
An install date is set	Pushes the appointment to the homeowner; queues 48 h and 24 h reminders. NOT SCHEDULED
A chat message is sent	Pushes to the app or emails; holds it through quiet hours (9pm–8am) and delivers in the morning NOT SCHEDULED ; sends the PM a twice-daily digest of unanswered messages NOT SCHEDULED ; never notifies anyone of their own message.
A PM asks the customer	Pushes an "action needed" notification; the answer lands back on the project.
Activity is logged	Updates the person's last-activity date; links the entry to project, deal and client.
A battery quantity is entered	Answers "System includes battery?" by itself, identically on screen and in reports.

Integrity guards (always on)

GUARD	EFFECT
Project hold on contact	A signed contact with a live project cannot leave Contract signed by any path.
Won deal read-only	A won deal's stage, contract value, client and dealer are fixed (admin excepted).
Deal on project immutable	The originating dealer and deal cannot be changed except by an admin with a reason.
Role guard	Only an admin can change anyone's role.
Audit log immutable	Log rows cannot be edited or deleted; 30 tables write row-level audit entries.
Guarded deletes	Dealers, customers and reference records with history cannot be deleted; lists deactivate instead.
Security	Account lockout after 10 failures; sessions revoked when a user is deactivated; upload links expire.

Computed live, never typed

Lifecycle (lead / prospect / customer / past customer) · days in every stage · total project days · deal probability (unless set by hand) and weighted pipeline · dashboard figures grouped by PM, dealer and stage · ageing against per-stage thresholds · the homeowner's plain-language status and typical duration · the missing-items count on every card.

Scheduled jobs

JOB	STATUS
<code>/api/push/reminders</code> : install reminders, quiet-hours flush, PM digest	NOT SCHEDULED Built, idempotent and protected by <code>CRON_SECRET</code> . Runs from Admin → Settings by hand; needs a cron entry (chapter 14).
Saved report schedules	NO RUNNER Schedules are stored; nothing executes them yet.

AI integration — current state

The product contains no calls to an AI model today. That was deliberate: phase one made the data right first, because AI working on wrong or scattered data is worse than no AI. The schema was designed from day one so that phase two needs no database redesign. Chapter 16 lists each AI capability to add; chapter 17 orders them into phases.

AI-ready foundations already in the database

TABLE	BUILT FOR	TODAY
price_book	Catalogue of parts and services with cost and price, for automated quoting and bills of materials.	Manual
adder_rules · project_adders	Rules that add line items (e.g. main panel upgrade) from a JSON condition; each adder records its <i>source</i> as <code>rule</code> , <code>manual</code> or <code>ai</code> .	Manual / rule
availability_slots	Designer and surveyor scheduling, so AI can book survey and review slots.	Manual
vendor_quotes · bom_items	Vendor pricing and bills of materials, for automated quote requests; source can be <code>ai</code> .	Manual
designs · design_assets · site_surveys	Structured design and survey data an AI designer or reviewer can read and write.	Manual
stage_feedback	Ratings and comments; source can be <code>ai</code> for AI self-critique of its own stage work.	Customer
exceptions	The work queue of things a person must look at: rule conflicts, low-confidence AI output , anything the automation will not decide alone.	Ready
permits · permit_events	Permit tracking with an event history, for automated status monitoring.	Manual

Why the platform is well placed for AI

Typed, structured stage data

Every stage field is a real column with a known type, so an AI's answer can be written into a column and validated, not pasted into a notes box.

Rules the AI cannot break

Stage validation, the hold, the won-deal lock and row-level security are all in the database. An AI agent acting through the same API gets exactly the same refusals as a person.

A place for doubt

The exceptions queue gives AI somewhere to put work it is not sure of, which is what lets it act alone on everything else.

A full audit trail

Every action is logged with actor and kind, so AI actions can be attributed, reviewed and reversed like anyone else's.

Work completed to date

85 commits between 3 August and 22 September 2026, grouped here by what they delivered. Each module shipped with its database migration, its screens and, from the CRM onwards, an automated test suite.

DATES	DELIVERED	DETAIL
3–5 Aug	Platform & pipeline	Plain-PostgreSQL platform (no Supabase): auth engine, file storage, row-level security, audit. Pipeline board, Projects tab, stage engine; sidebar shell with role-aware navigation; admin panel (users & roles, reference lists, settings, activity log); the six stage forms; Complete, Hold and Cancelled.
10–12 Aug	Projects	Complete column on the board; New Project form and Project Details with admin-managed equipment and financing lists; inverter and battery quantities.
14 Aug	Dealers, reports, customers	Dealer portal; dealer companies with commission defaults and guarded delete; drag-and-drop report builder; customer portal; customer management (search, merge, portal access, archive, anonymise). Health endpoint reporting applied migrations; re-runnable migrations; catch-up files; real database errors instead of generic 500s.
17 Aug	Mobile app	Installable customer app with push, camera and offline support; native Android and iOS projects built from the browser; homeowners sign in with a password; screens that degrade instead of breaking when the database is behind; self-diagnosing error pages.
19–21 Aug	Dashboard & chat	Full-width analytics dashboard; project chat with the internal-notes wall, global inbox, quiet hours and digest.
24–25 Aug	Experience	Three sign-in doors; portal redesigned around the current stage with typical durations; dark mode; stage feedback with email fallback, follow-ups and verbatim log; end-to-end sweep.
14–15 Sep	CRM foundation	Sales role; people with channels and addresses; dealer leads become deals; deals module with proposals and the Won handoff; the full contact intake; Create Contact and a Contacts list holding everybody.
17 Sep	Contacts & database	Contact fields separated from deal (solar) fields; Contact stages board; tooling to diagnose and catch up a database; then the application applying its own migrations from Admin → Database; contact record as a full page.
18 Sep	Stages & deployment	The eight business contact stages, full width, any-to-any; pill-button tabs; contact owner defaults to the creator; Lost in red; deployment fixes (per-branch Neon provisioning, TLS behaviour documented).

DATES	DELIVERED	DETAIL
22 Sep	Signing & deals	Contract signed opens the system form; signing creates the project and holds the contact; admin project delete; signing fix for databases with the first version; Deals board full width, then the project stages, live with Pipeline; sales reps see their deals' projects.

Quality work that shipped alongside

- **Test suites:** RLS matrix and auth flows (SQL); end-to-end suites for the CRM foundation, people, deals, contact intake, contact stages, migration apply, every "database behind" state, the signing upgrade path, and a real-browser suite for signing, the hold, deletion and live boards; 62 unit tests.
- **Diagnosability:** health endpoint, drift-aware screens, errors in PostgreSQL's own words, a branch fingerprint to confirm which database a deployment talks to.
- **Documentation:** README, auth design, RLS matrix, mobile store submission guide, deployment guide, and this document.

Known gaps & open items

Everything not finished, not switched on, or worth fixing, in priority order. Nothing on this list is hidden elsewhere in the document.

Do first: infrastructure & configuration

ITEM	WHAT TO DO	EFFORT
Scheduled jobs are not scheduled	Add a cron entry for <code>/api/push/reminders</code> and set <code>CRON_SECRET</code> . Note: on Vercel's Hobby plan a cron can run at most once a day; the reminders and quiet-hours flush want hourly, so use Vercel Pro, a GitHub Actions schedule, or an external scheduler calling the endpoint with the secret.	Hours
Neon ↔ Vercel integration	The integration's per-deployment branch step has failed deployments. Protect the branch holding the live data (set it as default), disconnect the integration, set <code>DATABASE_URL</code> (pooled, marked Sensitive) by hand, and rotate the database password that was exposed. See <code>docs/deployment.md</code> .	Hours
Confirm the live database branch	The data was found on a Vercel-created preview branch, not <code>main</code> . Confirm with the query in the deployment guide, and make that branch the production default before anything else changes.	Minutes
Backups	The Neon free plan keeps 6 hours of history. Move to a paid plan for days-long point-in-time restore, and add a scheduled off-site dump.	Hours
Environment variables	Confirm in Vercel: <code>SMTP_*</code> and <code>EMAIL_FROM</code> (all email), <code>VAPID_*</code> (push), <code>NEXT_PUBLIC_SITE_URL</code> (links in emails), <code>CRON_SECRET</code> . Without <code>SMTP</code> , invites, recovery and rating emails fail with a clear error.	Minutes

Functional gaps

ITEM	DETAIL	STATUS
Report schedules	Stored but not executed; needs a runner that emails the export on the timetable.	GAP
Subscribers (M26)	List and consent basis exist; the sender, double opt-in and preference page do not.	PARTIAL
Dealer CRM extensions	Module 18 of the CRM specification (dealer-side CRM features) not started.	PLANNED
CRM reports & sales band	Sales reporting and a sales band on the dashboard (lead-to-signed conversion, source ROI, rep leaderboards).	PLANNED
Pipeline board width	The Pipeline's nine columns overflow sideways even on wide screens; the Deals and Contact boards were fixed, Pipeline was not changed.	GAP
Returning customers	A completed project still holds its contact in Contract signed; a second sale needs the rule relaxed for complete/cancelled projects.	DECISION

ITEM	DETAIL	STATUS
Earlier signed contacts	Contacts signed before automatic project creation have no project; convert them from the Deals table.	ONE-OFF
SMS	The activity log knows the "sms" kind but nothing sends or receives SMS.	GAP
E-signature	Contract signed is recorded by a person; no DocuSign/PandaDoc-style capture or webhook.	GAP
External integrations	No connections yet to utilities, permitting portals, lenders, design tools, distributors or monitoring platforms.	GAP
Staff two-factor sign-in	Passwords with lockout only; admins and ops should have a second factor.	GAP
Continuous integration	The test suites run by hand; add a GitHub Actions workflow with a Postgres service to run them on every push. The browser suite needs <code>playwright-core</code> added as a dev dependency.	GAP
Placeholder addresses	Deals created before signing required an address may carry "Address to be confirmed"; signing now refuses it, but existing rows should be cleaned.	CLEANUP

Future automations — toward zero-touch

Rules-based automation that needs no AI: an event from outside, or a date, moves the work forward without anyone pressing a button. Each entry names what triggers it, what it does, and which manual step it removes.

Lead to signed contract

AUTOMATION	TRIGGER → ACTION	REMOVES
Web-to-contact capture	Website/landing-page form or ad platform webhook → Contact created with UTM source, medium, campaign and original source; duplicate check; owner assigned.	Re-typing web leads
Speed-to-lead response	New contact → within 60 seconds an SMS and email with a self-booking link; a call task if not booked in 15 minutes.	Delayed first contact
Round-robin & territory routing	New contact → owner chosen by ZIP code, language, rep capacity and availability.	Manual lead assignment
Self-booking	Homeowner picks a slot → Appointment scheduled, calendar invite to rep and homeowner.	Phone tag
Appointment reminders & no-show recovery	24 h and 2 h reminders with a reschedule link; a no-show moves the card to No-show and starts a rebooking sequence; rebooking moves it to Rescheduled.	Reminder calls, moving cards
Stage-based nurture	Quoted → follow-ups on days 1, 3, 7, 14; Lost → re-engagement at 90 days; each tied to the contact's stage.	Follow-up tracking
Financing status sync	Lender application link sent from the deal; lender webhook/API (e.g. GoodLeap, Mosaic, Sunlight, LightReach) → Financing approved, amount and terms written to the deal.	Checking lender portals
E-signature → signing	Contract sent from the deal; the e-signature platform's "completed" webhook → the existing signing step runs with the proposal's system, the signed PDF filed on the deal, and the project created.	The whole signing step
Proposal import	Design tool (e.g. Aurora, OpenSolar) proposal published → a new proposal version on the deal with system, price and production.	Copying proposal figures

Signed contract to permission to operate

AUTOMATION	TRIGGER → ACTION	REMOVES
Survey auto-scheduling	Project created → offer the homeowner open surveyor slots (<code>availability_slots</code>); confirmation books it and fills the survey date.	Scheduling calls
Evidence-based auto-advance	When every required field and photo of a stage is present and valid, the project advances on its own through the existing validation.	Pressing Advance
Design hand-off	Survey complete → design request sent to the designer queue or design partner; the partner's delivery webhook fills Designs Received and files the plan set.	Chasing designs
Permit & interconnection tracking	Portal notification emails and APIs (e.g. SolarAPP+ where the jurisdiction uses it, PowerClerk-style utility portals) → statuses and dates written to the permit tracks, with <code>permit_events</code> history.	Logging in to portals daily
Purchase orders	Permits approved → purchase order to the distributor from the bill of materials; shipment tracking → Material delivered.	Ordering, tracking
Install scheduling	Material delivered → the job is placed on the crew calendar by capacity, travel and weather; homeowner confirms in the app; day-of ETA message.	Crew scheduling
Inspection scheduling	Install complete → inspection request to the jurisdiction; a failure opens a correction task from the failure notes.	Follow-up admin
Monitoring & energization	Monitoring platform API (e.g. Enphase, SolarEdge, Tesla) reports first production → energization date filled, "power on" sent, production checked at 7 and 30 days.	Manual activation checks
Delay alerts	A stage passes its ageing threshold → the PM and manager are alerted and the homeowner gets a proactive "this is taking longer" note.	Surprise escalations

Money, reviews and the business

AUTOMATION	TRIGGER → ACTION	REMOVES
Milestone invoicing	Stage complete → invoice for the matching payment milestone; payment platform/accounting sync (e.g. Stripe, QuickBooks) → milestone status Received.	Billing and reconciling
Finance milestone submission	Milestone reached → evidence package sent to the finance partner; approval updates Finance M1/M2.	Lender paperwork
Dealer commissions	Milestones paid → commission statements calculated from the dealer's defaults, sent monthly.	Commission spreadsheets

AUTOMATION	TRIGGER → ACTION	REMOVES
Reviews & referrals	Completion with a high rating → review request and referral invitation; a low rating goes to follow-up instead.	Asking for reviews
Scheduled reports	A runner for the saved report schedules → Excel export emailed to recipients.	Weekly report building
Homeowner weekly digest	Weekly → a one-screen status summary to every active homeowner.	Status calls
Data hygiene	Nightly → duplicate-merge suggestions, stale deals flagged, missing next actions listed per rep.	Cleanup days
Operations watch	Health endpoint monitored every few minutes; failures and error spikes alert the admin.	Finding outages from users

Future AI integrations

Where judgement, reading or language is involved and rules alone cannot finish the job. For each capability: what the AI does, where its answer goes, how far it can act alone, and where a person stays in the loop.

Recommended model approach

Use a hosted model family through one server-side integration, for example the Anthropic Claude API: a capable general model (e.g. Claude Sonnet 5) for drafting, reasoning and document reading, a fast low-cost model (e.g. Claude Haiku 4.5) for high-volume classification and routing, and vision input for photos and scanned documents. Pair it with solar data services: Google Solar API or aerial imagery providers (e.g. EagleView, Nearmap) for roof geometry and sunlight, and NREL PVWatts for production estimates. Every AI result is written through the same API and database rules as a person's.

Sales & customer

CAPABILITY	WHAT THE AI DOES → WHERE IT LANDS	AUTONOMY TARGET
Intake assistant	Reads inbound emails, web messages and call transcripts → fills contact fields, detects duplicates, sets source.	Acts; flags doubts
Lead scoring	Roof suitability (imagery + sunlight), homeownership, utility rate, bill size, engagement → priority score and routing.	Acts
AI appointment setter	Two-way SMS (and optionally voice) that answers questions, books, confirms and reschedules within consent rules; hands over to a rep on request or complexity.	Acts; escalates
Utility-bill reader	Photo or PDF of the bill → utility, rate plan, monthly and annual kWh, average bill: written to the deal's typed usage fields.	Acts; low confidence → exceptions
Instant preliminary design	Address + usage → roof segments, module count, system size, estimated production and offset → a draft proposal version.	Drafts; rep approves
Quote & adders	Survey facts + price book + adder rules → line items (e.g. main panel upgrade) with <code>source = 'ai'</code> .	Drafts; approval over threshold
Homeowner copilot	Answers status questions 24/7 in chat from the plain-language status and typical durations; drafts PM replies for anything else; always discloses it is automated.	Acts on routine; drafts the rest
Conversation summaries	Calls, chats and emails → a short summary and next action in the activity log.	Acts
Cancellation-risk signal	Ratings, chat sentiment, delays and response times → a risk score that triggers proactive outreach before a cancellation.	Alerts

Delivery

CAPABILITY	WHAT THE AI DOES → WHERE IT LANDS	AUTONOMY TARGET
Survey photo QA	Checks every required shot (roof planes, attic, main panel, meter, service drop) is present and legible; spots panel-upgrade or roof-condition issues → adders and a survey pass/fail.	Acts; failures to PM
Document classification	Every upload is recognised and filed in the right category (bill front/back, panel, meter, agreement, plan set, permit).	Acts
Plan-set pre-check	Reads the plan set against the jurisdiction's requirements and common code items (e.g. NEC 690/705, rapid shutdown, fire setbacks, labelling) before submission → a list of likely rejections.	Advises; engineer decides
Permit packet assembly	Collects plans, spec sheets, forms and signatures required by the jurisdiction and utility into a submission-ready packet.	Drafts; PM submits → later acts
Portal & inbox monitor	Reads permit, interconnection and utility emails and portals → updates statuses, dates and revision notes; drafts responses to revision requests.	Acts; revisions to PM
Install photo QA	Checks install pictures for compliance items (labels, conduit, rapid-shutdown devices, grounding, workmanship) before inspection is requested.	Acts; failures to install manager
Inspection-failure triage	Failure notes → corrective task list with parts and time estimate.	Drafts
Delay prediction & forecasting	Stage history → which projects will slip, and the install and revenue forecast for the next 90 days.	Advises

Back office

CAPABILITY	WHAT THE AI DOES → WHERE IT LANDS	AUTONOMY TARGET
Ask the data	"Installs by dealer last quarter" → a report definition for the existing builder. Safe by design: the builder takes JSON over whitelisted fields, never SQL.	Acts
Commission & payment checks	Flags commissions, milestones and invoices that do not match contract terms.	Alerts
Duplicate & data-quality review	Suggests merges and corrections with a confidence score; high-confidence cases merge automatically.	Acts above threshold
Self-review	The AI rates its own stage work into <code>stage_feedback</code> with <code>source = 'ai'</code> , so quality can be tracked over time.	Acts

Guardrails for every AI feature

Confidence and the exceptions queue

Every AI action carries a confidence. Below the threshold it goes to **exceptions** for a person; above it, it acts. Thresholds are per capability and tuned from real outcomes.

People for money, law and safety

Prices over a limit, contract terms, engineering sign-off, electrical safety and anything a regulator requires a licensed person for always stay with a person.

Consent and disclosure

AI texts and calls only with prior express written consent, within quiet hours, honouring do-not-call; the AI identifies itself. Consent basis is stored (as M26 already does).

Measured, logged, reversible

Every AI action is in the audit log under an AI actor. Each capability has a test set it must pass before release and a dashboard of accuracy and cost.

Autonomy roadmap

A realistic path from today to a pipeline where people handle only exceptions and the work that law and physics reserve for them.

Levels of autonomy

LEVEL	NAME	MEANING
L0	Manual	A person does the work; the system records it. (Most stage data entry today.)
L1	Assisted	The system checks and blocks mistakes. (Stage validation, duplicate guard, hold: today.)
L2	Drafted	AI prepares the work; a person approves with one click.
L3	Supervised	AI or rules act on their own; people review exceptions and samples.
L4	Autonomous	The step runs end to end; people see results and outliers only.

Phases

PHASE	WHEN	SCOPE
0 · Switch on	Weeks 0–2	Schedule the reminders job; fix the database hosting (integration, default branch, backups, rotated password); confirm email and push configuration; add CI for the test suites; fit the Pipeline board to the screen.
1 · Connect	Months 1–2	SMS provider; self-booking calendar; speed-to-lead sequence; e-signature webhook into signing; lender status sync; monitoring platform API; report schedule runner; staff two-factor sign-in.
2 · AI drafts (L2)	Months 2–4	Utility-bill reader; document classification; conversation summaries; homeowner copilot drafting replies; survey and install photo QA flags; preliminary design drafts.
3 · AI acts (L3)	Months 4–8	Copilot answers routine status questions itself; permit/interconnection monitoring writes statuses; survey auto-scheduling; evidence-based auto-advance; adders and quotes below a price limit; cancellation-risk outreach.
4 · Near zero-touch (L4)	Months 8–12	AI appointment setter; permit packet submission where portals allow; purchase orders; install scheduling optimisation; forecasting; auto-merge of high-confidence duplicates.

What stays with people

"No human interaction" is achievable for the office work: data entry, chasing, scheduling, status updates, filing and reporting. It is not achievable, and should not be attempted, for the steps that require a licensed or accountable person: the homeowner's signature and financing consent; a licensed engineer's stamp; the physical survey and installation; the jurisdiction's inspection; the utility's permission to operate; and final decisions on pricing exceptions, cancellations and refunds. The target is that people spend their time only on those steps and on the exceptions queue.

Best practices — solar operations

Practices that consistently separate installers with short cycle times and few cancellations from the rest. Where the platform already supports a practice, it says so; where it could, that is noted.

Selling the right system

- **Size from twelve months of usage**, not one bill. Read the full year (the utility-bill reader in chapter 16 makes this effortless) and size for the target offset under the utility's current export rules.
- **Understand the export tariff**. Where net metering has been replaced by lower export credits (for example California's net billing tariff), self-consumption matters more and batteries become part of the value, not an add-on.
- **Estimate production honestly**. Use a recognised model (PVWatts or your design tool) with real shading, and quote a central (P50) figure. Overpromised savings are the leading source of complaints and cancellations.
- **Check the roof and the panel before the contract**. A roof with little life left should be replaced first; a main-panel upgrade discovered at survey is the most common surprise adder. Capture both on the deal before signing.
- **Check the HOA and the utility early**. HOA approval and interconnection limits can each stop a job; the permit tracks already hold both, so start them at signing, not after design.
- **Treat incentives as time-sensitive facts**. The US residential clean energy credit (IRC §25D) ended for systems paid for after 31 December 2025, while third-party-owned systems (lease/PPA) claim business credits with their own deadlines and conditions. State and utility programmes change often. Verify with a tax professional before quoting savings, and date every proposal.
- **Disclose clearly**. Price, financing terms, dealer fees, escalators, production assumptions and the buyer's cancellation right. For sales made in the home, the federal three-day cooling-off rule applies, and many states add more.

Running the delivery pipeline

Survey

- A fixed photo checklist per survey (every roof plane, attic/rafters, main panel with the cover off, meter, service entrance, planned equipment locations).
- Measure, do not estimate; record roof age and material.
- Survey within days of signing: the gap is where buyer's remorse grows.

Design

- Design against the specific jurisdiction's and utility's rules (setbacks, rapid shutdown, labelling, interconnection method).
- Run a pre-submission checklist to cut revision cycles; track revisions per project.
- Get engineering stamps in parallel where possible.

Permits & interconnection

- Maintain a requirements sheet per jurisdiction and utility.
- Use instant online permitting (e.g. SolarAPP+) where the jurisdiction has adopted it.
- Submit interconnection as early as the utility allows, in parallel with the building permit.
- Measure first-time approval rate and chase anything past its typical duration.

Procurement

- An approved equipment list with alternates ("okay to install comparable brand" is already asked at signing).
- Confirm stock before scheduling installs; stage material per job.
- Watch tariff and domestic-content rules that affect cost and incentives.

Install

- A pre-install call and a day-of arrival message.
- Safety first: fall protection and electrical safety procedures, every job.
- A standard photo set before leaving site (array, attachments, conduit runs, labels, rapid-shutdown devices, grounding, inverter/battery, main panel) — the install-photo QA in chapter 16 checks it.

Inspection, PTO & after

- Self-inspect from the photo set before booking the inspector; track first-time pass rate.
- Request PTO immediately after passing; activate monitoring at energization.
- Check production against the estimate at 7 and 30 days; resolve underperformance before the customer notices.

Keeping the homeowner

- **One named project manager** and one channel (the platform's chat) for the life of the job.
- **Tell them before they ask:** proactive updates at every stage change, typical durations on screen, and a note when something runs long.
- **Measure every stage** (the stage ratings) and act on every low score within a day.
- **Ask for the review and the referral at the moment of delight:** power on, and a high final rating.

Best practices — CRM & sales

Speed and persistence

- **Respond in minutes, not hours.** Contact rates fall steeply after the first five minutes; an automated instant response plus a quick human call is the single biggest lever on close rate.
- **Use a cadence:** calls, texts and emails over one to two weeks, then a longer nurture.
- **Protect the appointment:** confirmation at booking, reminders the day before and two hours before, and an easy reschedule link. A rescheduled appointment is far better than a no-show.

Data you can trust

- **One system of record.** No side spreadsheets; if it is not in the CRM it did not happen.
- **Minimum fields, enforced; everything else optional.** The platform requires a surname and a way to reach the person, and no more, so people do not invent data to get past the form.
- **Stop duplicates at entry** (the platform checks every email and phone on file) and merge the rest promptly.
- **Attribution from the first touch:** source, campaign and dealer on every contact, so cost per sale by source is measurable.

Pipeline discipline

- **Every stage has exit criteria** that the system checks (the deal stages already refuse to advance without them).
- **Every open deal has a next action and a date** (enforced), and a weekly pipeline review walks through anything overdue.
- **A reason for every loss,** from a fixed list (enforced), reviewed monthly; lost is reversible, so re-engage at intervals.
- **Visibility by role and territory,** with an unassigned pool so nothing waits for a manager to hand it out.

Consent and compliance

- **Calls and texts:** automated or AI-generated calls and texts to mobiles need prior express written consent under the US TCPA (the FCC treats AI voices as artificial voices). Honour do-not-call lists, contact only 8am–9pm local, and check stricter state rules.
- **Email and privacy:** CAN-SPAM for commercial email; double opt-in with the consent basis stored (M26 shows it); honour deletion requests (the anonymise function exists for this).

Dealers and partners

Agree lead-acceptance rules and response times in both directions; send each dealer a monthly scorecard (leads, appointment rate, close rate, cancellation rate, cycle time, customer rating); and make commission statements automatic and transparent, which prevents more disputes than any policy document.

Best practices — security, data & operations

AREA	PRACTICE	PLATFORM
Secrets	Store connection strings and keys as Sensitive variables; never paste them into chat, email or tickets; rotate at once if exposed.	ACTION
Backups	Point-in-time restore measured in days, plus a scheduled off-site dump; test a restore quarterly.	UPGRADE
Environments	Production on the default database branch; a separate staging database for trying migrations first.	ACTION
Migrations	Never edit a shipped migration; apply through Admin → Database; keep every file idempotent and guarded.	IN PLACE
Access	Least privilege by role, enforced in the database; deactivate leavers the same day; review admin accounts quarterly.	IN PLACE
Sign-in	Two-factor for admin and ops accounts; lockout on repeated failures (in place).	2FA TO ADD
Audit	Immutable activity log; review it monthly for unusual role changes, deletions and exports.	IN PLACE
Personal data	Utility bills, account numbers and addresses are sensitive; limit who can see documents, and anonymise on request.	IN PLACE
Monitoring	Uptime check on <code>/api/health</code> ; alerting on server errors; watch database connections and storage size.	TO ADD
Testing	Run the end-to-end and SQL suites automatically on every push; block merges on failure.	CI TO ADD
File storage	Files in Postgres suit today's volume; plan a move to object storage when stored files pass tens of gigabytes. The metadata is already shaped for it.	LATER
Mobile	Keep store signing keys in a password manager and in GitHub secrets only; never pin the app to a deployment URL (the build refuses).	IN PLACE

KPIs & targets

Measure a baseline for three months before setting targets; industry figures vary widely by market and channel. "Where" says whether the platform can already report it.

Sales

KPI	DEFINITION	WHERE
Speed to lead	Minutes from contact created to first two-way contact.	Activity log · report to build
Appointment set rate	Contacts scheduled ÷ contacts created.	Contact stages
Show rate	Appointments held ÷ scheduled.	Contact stages
Close rate	Contract signed ÷ appointments held.	Contact stages · deals
Cancellation rate	Signed projects cancelled before PTO ÷ signed.	Pipeline · dashboard
Average deal	kW and value per signed deal; battery attach rate.	Deals · reports
Cost per sale by source	Marketing spend ÷ signed deals, by source.	Needs spend data

Delivery

KPI	DEFINITION	WHERE
Sale to install	Days from contract signed to install complete.	Stage history · dashboard
Install to PTO	Days from install complete to permission to operate.	Stage forms
Days per stage vs typical	Each stage against its typical range and threshold.	Dashboard
First-time permit approval	Approved with no revision ÷ submitted.	Permit tracks · report
First-time inspection pass	Inspections passed first time ÷ inspections.	Inspection status · report
On-hold share	On Hold ÷ active; days held, by reason.	Pipeline · dashboard

Customer & money

KPI	DEFINITION	WHERE
Stage rating	Average score per stage, and share of low scores.	Feedback · dashboard
Follow-up closure time	Hours from low score to follow-up closed.	Follow-ups
Chat first response	Minutes to first PM reply during working hours.	Messages · report to build
Review & referral rate	Reviews and referrals per completed project.	Needs tracking
Milestone collection	Days from milestone reached to payment received.	Payment milestones · report

Database migrations

Every schema file, in the order it is applied. Generated from `db/migrations` when this document is built.

#	NO.	FILE	WHAT IT DOES
1	000000	platform	Platform baseline (plain PostgreSQL, no Supabase)
2	000100	init_schema_and_enums	Init: app schema, enums, generic helpers
3	000200	tables	All tables (§3), incl. AI-era tables created now so phase two needs
4	000300	access_helpers	Access helpers used by every RLS policy (000600, 000700)
5	000400	hooks_and_views	User lifecycle, JWT claim hook, stage-history trigger, finance view
6	000500	audit	Audit log writer (the shared utility every later module calls)
7	000600	rls_policies	Grants + Row-Level Security for every §2 role
8	000700	storage	Storage: private buckets for DWG / PDF deliverables / photos
9	000800	add_ops_role	Add the Ops/PM role (§2 addendum, auth module)
10	000900	auth_module	Auth module: ops access rules, upload grants (REQ-SEC-01),
11	001000	auth_engine	Auth engine (replaces Supabase Auth)
12	001100	file_storage	File bytes in Postgres (replaces Supabase Storage)
13	001200	manual_version	Manual version (SolarFlow PM): six stages, stage data model,
14	001300	admin_panel	Admin panel: user management engine, reference-record fields,
15	001400	stage_fields	Stage field specification (data-entry forms)
16	001500	complete_hold_cancel	Stage 7 (Complete) + Hold & Cancelled side stages
17	001600	complete_stage_backfill	Backfill: completed projects land on the Complete stage
18	001700	project_details	New Project form & Project Details (equipment + financing lists)
19	001800	equipment_quantities	Inverter & battery quantities
20	001900	dealer_portal	Dealer Portal (leads, commissions, per-field dealer visibility)
21	002000	dealer_companies	Dealer companies (admin record, commission defaults, guarded delete)
22	002100	restore_project_defaults	Re-assert the projects defaults
23	002200	report_builder	Report builder (saved reports, schedules, run history, indexes)
24	002300	customer_portal	Customer portal (plain-language mapping, requests, PM estimate)
25	002400	customer_management	Managing customers (identity, merge, archive, anonymise)
26	002500	mobile_app	Customer mobile app (installable portal, push, PM asks)
27	002600	customer_passwords	Homeowners sign in with a password
28	002700	invite_customers_with_tokens	Auto-invited homeowners get a set-password link
29	002800	dashboard	Analytics dashboard
30	002900	project_chat	Project chat (PM ↔ customer)

#	NO.	FILE	WHAT IT DOES
31	003000	sign_in	Sign-in & role entry
32	003100	typical_durations	Typical stage durations (customer portal redesign §5, §7)
33	003200	stage_feedback	Stage feedback (customer sentiment)
34	003300	add_sales_role	Add the Sales role (Modules 16–19, Part 8)
35	003400	crm_foundation	Part 2 and Part 10 — the CRM foundation
36	003500	deals	Part 5 and Part 9 — deals, proposals and the Won handoff
37	003600	contact_intake	the contact intake fields
38	003700	contact_create	creating a contact
39	003800	contact_stages	Contact stages — the contact's own pipeline
40	003900	contract_signed_system	Contract signed — the system is recorded at the moment it is sold
41	004000	project_holds_contact	The project holds the contact in place, until it is deleted
42	004100	signing_creates_project	Signing creates the project
43	004200	sales_see_deal_projects	A sales rep sees a deal's project exactly when they can see the deal

Screens & API endpoints

Generated from `src/app` when this document is built.

Screens

AREA	ROUTES
(home)	/
admin	/admin · /admin/[entity] · /admin/activity · /admin/customers · /admin/database · /admin/dealers · /admin/finance · /admin/people · /admin/people/[id] · /admin/people/new · /admin/people/stages · /admin/settings · /admin/subscribers
auth	/auth/change-password · /auth/update-password
dashboard	/dashboard
dealers	/dealers · /dealers/commissions · /dealers/leads · /dealers/login · /dealers/projects · /dealers/projects/[id]
deals	/deals · /deals/[id] · /deals/new
designer	/designer
feedback	/feedback
leads	/leads
login	/login · /login/dealer · /login/homeowner · /login/reset
messages	/messages
pipeline	/pipeline
portal	/portal · /portal/documents · /portal/login · /portal/messages · /portal/more · /portal/photos · /portal/project
projects	/projects · /projects/[id] · /projects/[id]/chat · /projects/[id]/chat/print · /projects/[id]/stages/[stage] · /projects/new
r	/r/[token]
reports	/reports · /reports/builder
tasks	/tasks
u	/u/[token]

API endpoints

GROUP	ROUTES
admin	/api/admin/dealers · /api/admin/dealers/[id] · /api/admin/migrations · /api/admin/records/[entity] · /api/admin/settings · /api/admin/users · /api/admin/users/[id]
app	/api/app/version
auth	/api/auth/change-password · /api/auth/login · /api/auth/password · /api/auth/recovery
chat	/api/chat/[projectId] · /api/chat/[projectId]/flag · /api/chat/[projectId]/messages/[messageId] · /api/chat/[projectId]/read
contacts	/api/contacts · /api/contacts/[id]/sign · /api/contacts/[id]/stage

GROUP	ROUTES
customers	/api/customers · /api/customers/[id] · /api/customers/[id]/detail · /api/customers/[id]/intake · /api/customers/[id]/portal · /api/customers/lookup · /api/customers/merge
deals	/api/deals · /api/deals/[id]/activity · /api/deals/[id]/documents · /api/deals/[id]/move
documents	/api/documents/[id]
feedback	/api/feedback/[projectId]/[action]
files	/api/files/[id]
health	/api/health · /api/health/probe
invites	/api/invites
leads	/api/leads · /api/leads/[id]
portal	/api/portal/delete-account · /api/portal/documents/zip · /api/portal/requests · /api/portal/uploads
projects	/api/projects · /api/projects/[id] · /api/projects/[id]/asks · /api/projects/[id]/commission · /api/projects/[id]/details · /api/projects/[id]/documents · /api/projects/[id]/move · /api/projects/[id]/requests/[requestId] · /api/projects/[id]/stages/[stage] · /api/projects/bulk
push	/api/push/key · /api/push/prefs · /api/push/reminders · /api/push/subscribe
r	/api/r/detail
refs	/api/refs/[entity]
reports	/api/reports · /api/reports/[id] · /api/reports/export · /api/reports/preview
tasks	/api/tasks/[id]/resolve
u	/api/u/[token]

Tests, configuration & tooling

Automated tests

SUITE	PROVES
db/tests/rls_verification.sql	The role-access matrix: each role sees exactly its rows.
db/tests/auth_flows.sql	Sessions, logout, one-time codes, invites, recovery, upload-grant expiry and revocation.
npm run test:unit	62 unit tests: route access truth table, drift classification and more.
e2e/crm-foundation.sh	The CRM scripts apply over live data; channels, lifecycle, leads becoming deals, re-runnable.
e2e/people.sh	People screen and contact page; lifecycle filter; duplicate detection on any channel; web enquiries with no invented attribution; deletion and suppression of subscription-only prospects; one activity log.
e2e/contact-intake.sh	The contact form's exact fields; one save split between person and deal; documents on the deal; the deal picker; Create Contact and its required rules.
e2e/contact-stages.sh	The eight stages, any-to-any moves, signing, the project hold, admin deletion.
e2e/deals.sh	Deal rules, proposals, Won creates the project, the live board, sales visibility.
e2e/apply-migrations.sh	Admin → Database applies missing files, admin-only, idempotent, reports refusals.
e2e/schema-behind.sh	Every screen loads with the database stopped at each of 13 earlier states.
e2e/signing-upgrade.sh	A database with the first signing function is detected and fixed by Apply.
e2e/contract-signed-ui.sh	In a real browser: the signing popup, the hold, deletion, and live Deals/Pipeline updates.

Environment variables

VARIABLE	PURPOSE
DATABASE_URL	Pooled PostgreSQL connection string (store as Sensitive).
DATABASE_SSL	Only for connection strings without <code>sslmode</code> ; not needed with Neon.
NEXT_PUBLIC_SITE_URL	The public address used in emailed links.
SMTP_HOST · SMTP_PORT · SMTP_USER · SMTP_PASS · SMTP_SECURE · EMAIL_FROM	Outgoing email.
VAPID_PUBLIC_KEY · VAPID_PRIVATE_KEY · VAPID_SUBJECT	Web push (generate with <code>npm run make:vapid</code>).
CRON_SECRET	Authorises the scheduled reminders endpoint.
PRIVACY_REQUEST_EMAIL	Where account-deletion requests are sent.
EMAIL_DEV_LOG	Development only: log emails instead of sending.

Commands

COMMAND	DOES
<code>npm run db:migrate</code>	Apply migrations from a terminal (Admin → Database does the same from the browser).
<code>npm run db:create-admin</code>	Create the first admin account.
<code>npm run db:demo-customer</code>	Create a demonstration homeowner and project.
<code>npm run db:verify</code>	Run the SQL verification suites on a throwaway database.
<code>npm run make:icons · make:vapid</code>	App icons; push keys.
GitHub Actions: Build Android app / Build iOS app	Mobile builds from the browser (debug APK or signed store bundle).

Glossary

TERM	MEANING
Adder	An extra line item on a solar job (e.g. main panel upgrade, trenching, roof work).
AHJ	Authority having jurisdiction: the city or county that issues the permit and inspects.
Contact stage	Where a person stands in the sale (created → signed / lost). Belongs to the contact.
Deal stage	The opportunity's own stages (New → Contract out → Won / Lost).
Drive Updated	The gate closing each stage: the job's shared drive has the stage's documents.
Energization	The system switched on and producing; monitoring active.
HDM NTP	Notice to proceed from the finance partner before installation.
HOA	Homeowners' association approval, where the property has one.
ICA	Interconnection agreement with the utility.
kW / kWh	System size (kilowatts, DC) / energy produced or used (kilowatt-hours).
Lifecycle	Lead, prospect, customer, past customer: derived from the person's projects.
Migration	An ordered SQL file that changes the database schema.
Offset	The share of annual usage the system is expected to produce.
PM / Ops	Project manager: the role that runs the pipeline.
PPA / lease	Third-party ownership: the homeowner pays for power or rent, not the system.
PTO	Permission to operate, granted by the utility after inspection.
Rapid shutdown	The electrical code requirement to de-energize rooftop conductors quickly for firefighter safety.
RLS	Row-level security: database rules that decide which rows each user may read or change.
Stamps	A licensed engineer's approval on the plan set.
The hold	The rule that keeps a signed contact in Contract signed while their project exists.

This document was generated from the repository at commit 725548b on 22 September 2026. Its source is `docs/project-documentation/`; rebuild it with `node docs/project-documentation/build.mjs` as the platform changes.